

# Can a short prefix choose a useful cache policy?

Seven public traces. A controlled synthetic agent overlay. An experiment you can inspect and refit.

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## EXPERIMENT DESIGN

# Hold out the whole trace.

A small tree chooses among WTinyLFU, S3FIFO, and GDSF. All selected policies use a rate guard.

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**7 × 3 × 3**

Traces × overlay behaviors × target agent fractions

63 related settings, seven whole-trace holdout groups

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**1,000**

Mixed requests in the selector prefix

Four candidate statistics; at most two in each fitted tree

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**64 MiB**

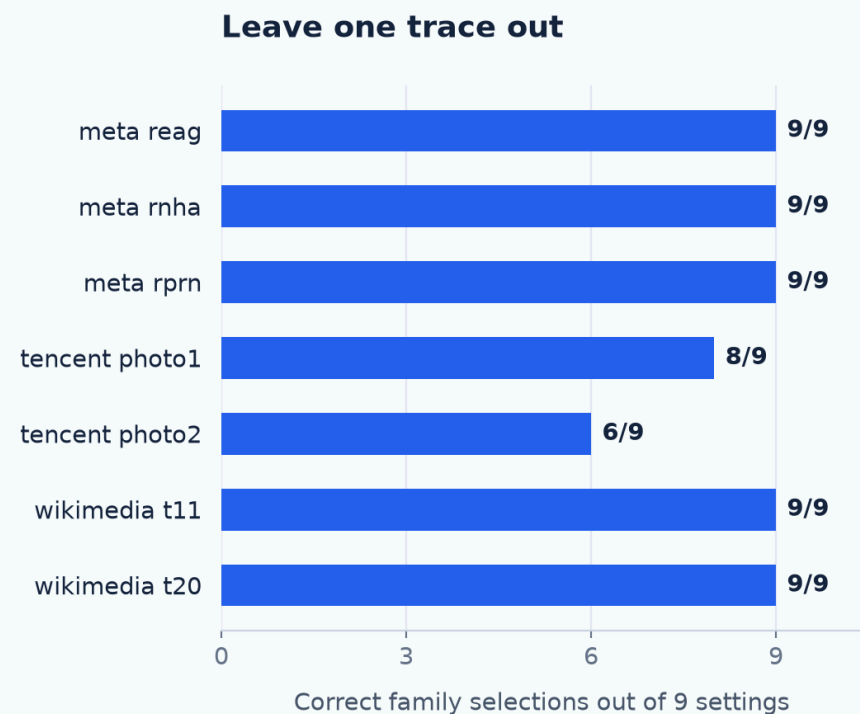
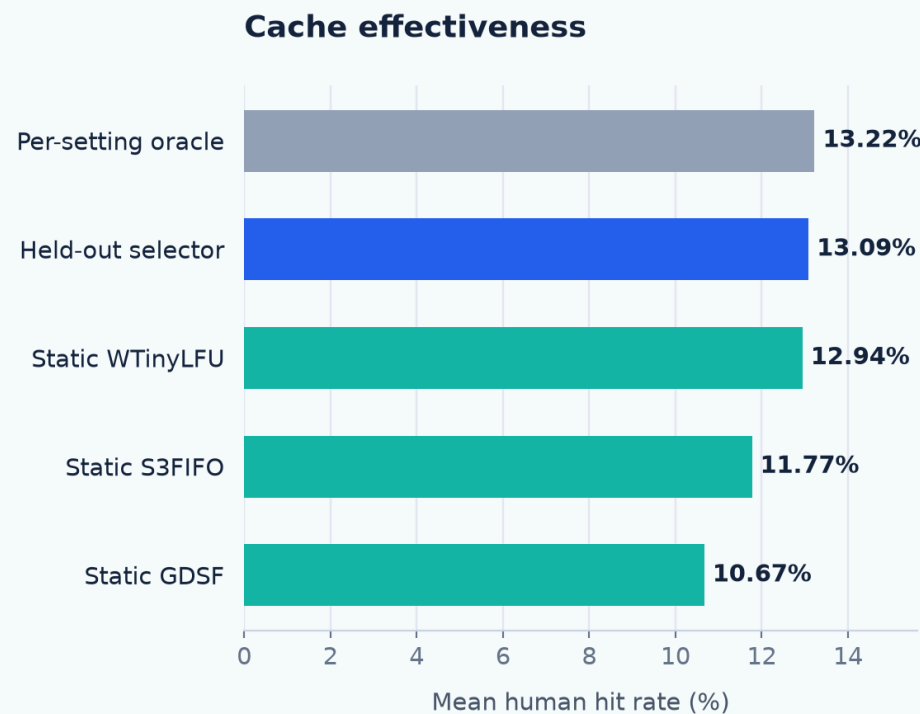
Fixed, stressed cache budget

513 archived policy cells; policy menus differ across trace panels

# 59 / 63 correct family selections

## CacheRegime

59 / 63 correct family choices · 7 public traces · 64 MiB cache



Controlled WebLINX-derived synthetic agent overlay; three behaviors × three agent fractions per trace.  
Human hit rates are means across 63 settings. Selector and static baselines use rate guards; oracle may choose any measured variant.  
Source: results/summary.json · Refit from frozen features; full raw-trace simulation is outside this artifact.

## INTERPRETATION

# A modest gain, with visible limits.

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## +0.141 pp

Mean human hit-rate gain over the strongest fixed baseline

Unweighted mean over settings; not a pooled production estimate

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## 4 errors

All four occur on the two Tencent traces

Five other traces score 9/9 each

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## No transfer claim

Collected-agent follow-up changes the winning families

Different features and workload construction prevent a matched comparison

## REPRODUCIBILITY

# Read the rows. Refit the folds.

```
python3 scripts/reproduce.py
```

Checks source hashes, reconstructs 63 oracle labels from 513 cells, refits all seven folds, and compares every prediction with the archived record.

Python standard library only. Frozen aggregate inputs and prefix features. Full raw-trace simulation is a separate reproduction stage.

[Repository overview](#) · [Technical note](#) · [Input hashes](#)